Summary in Graph

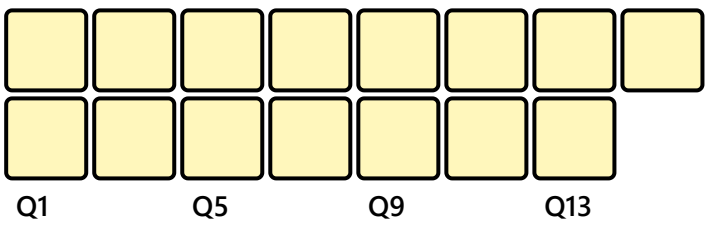
Exam Summary (GO Classes GATE CS | Computer Networks | Weekly Quiz 6)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (19) out of 189 users

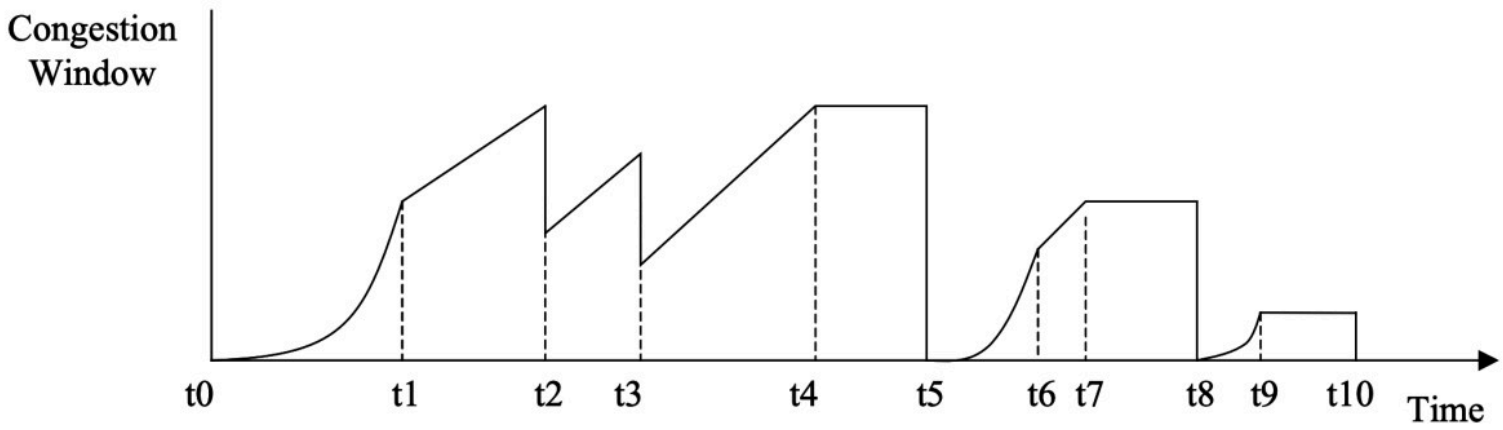
EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Computer Networks

The figure below illustrates the variation in the congestion window size of a TCP connection over time:

(On receiving 3 duplicate ACKs, the congestion window is reduced to half.)



Which of the following statements is/are correct?

A. TCP is in the slow start phase during the time periods $t_0 - t_1$, $t_5 - t_6$, and $t_8 - t_9$.
B. TCP is in the congestion avoidance phase during the time periods $t_0 - t_1$, $t_5 - t_6$, and $t_8 - t_{10}$.
C. TCP receives duplicate ACKs at time points t_2 and t_3.
D. TCP experiences timeouts at time points t_2, t_3, and t_9.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

For a client-server application running over TCP or UDP, which of the following is true

A. the server program must be executed before the client program

B. the client program must be executed before the server program

C. the server program can be executed before or after the client program

D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Host A sends two TCP segments to Host B back-to-back. Host B has already received bytes up to 110. The first segment sent contains 100 bytes with a sequence number 111, and the second contains 10 bytes. If the second segment arrives before the first, what will be the acknowledgment number sent by Host B in response?

A. 111

B. 211

C. 121

D. 110

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

In TCP, define cwnd to be the congestion window size and rwnd to be the receive window size; the amount of unacknowledged data can not exceed

A. rwnd

B. cwnd

C. min{cwnd, rwnd}

D. max{cwnd, rwnd}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Professor Rao is explaining TCP congestion control to his students. He presents the following scenario:

- A TCP sender is operating in the slow start phase.
- The Maximum Segment Size (MSS) $= 1000$ bytes.
- Immediately after the 10th ACK, the Congestion Window (CWND) is 2000 bytes.
- With this window, the sender transmits packets 11 and 12.
- Now, the ACK for packet 11 arrives.

According to TCP slow start rules, which new packets will the sender transmit at this point?

A. Packet 13 only

B. Packets 13 and 14

C. Packets $13, 14, 15$

D. None (the sender must wait for more ACKs)

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

If a TCP sender has $14,600$ bytes of data in-flight (i.e., it has sent those bytes but has not yet received acknowledgment for any of them), which of these statements is NOT NECESSARILY true?

A. The congestion window size of the sender is at least 10 MSS's (or, $14,600$ bytes)

B. The congestion window size of the receiver is at least 10 MSS's (or, $14,600$ bytes)

C. The sender is expecting acknowledgment packets from the receiver

D. The flow control window advertised by the receiver to the sender is at least $14,600$ bytes

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

You are watching a GO Classes Computer Networks class in one browser tab, and you also open another tab to browse a cat-related blog. Both tabs establish TCP connections, and your laptop has only one IP address.

When a TCP segment arrives, how does the operating system decide whether it belongs to the GO Classes lecture or to the cat blog, ensuring that packets don't get mixed up?

A. By checking only the IP address, since both tabs are on the same machine.

B. By using the port numbers in the TCP header, which let the OS map each segment to the correct socket/application.

C. By analyzing the application-layer data inside the payload.

D. By ordering packets based on their arrival timing.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Bob's TCP stack has received data up to byte 560 from Alice, but Bob's browser has only read 60 bytes. Bob's buffer size is 1024 bytes. What will be the advertised window size in the next TCP header that Bob sends?

A. 464

B. 524

C. 1024

D. 500

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose that the advertised receiver window in TCP is 1 Mbyte long. If a sequence number is chosen uniformly at random from the entire TCP sequence number space (size $= 2^{32}$), what is the probability that the sequence number falls inside the advertised window?

A. $\frac{1}{2^{32}}$

B. $\frac{1}{2^{12}}$

C. $\frac{1}{2^{20}}$

D. $\frac{1}{2^{16}}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

At some point, a TCP connection has the following state:

- CWND = 4 MSS, SSTHRESH = 8 MSS
- All packets up to and including $\#411$ have been sent and acknowledged.
- Packets $\#412$ through $\#415$ have been sent but not yet acknowledged.
- Packets $\#416$ and later have not been sent.

Now, assume TCP is in Congestion Avoidance mode.

If an ACK for packet $\#412$is received, which packet(s) will be sent as a result? (Select all that apply.)

A. $\#416$

B. $\#417$

C. $\#418$

D. $\#419$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A TCP sender has MSS = 100 B and a constant window size = 400 B.

- p_x: a packet whose first byte has the sequence number x.
- t_x: time at which packet p_x is transmitted.

At time t_{1600}, packet p_{1600} (i.e., the packet containing bytes starting from 1600) has just been transmitted. What is the highest ACK received by the sender by this time?

A. 1000

B. 1200

C. 1300

D. 1600

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Alice is sending data to Bob over a TCP (Transmission Control Protocol) connection in the congestion avoidance (Additive Increase, Multiplicative Decrease) phase.

- Maximum Segment Size (MSS) = 100 bytes
- Current Congestion Window (CWND) $= 1000$ bytes
- Alice sends a packet with sequence number 801, and Bob replies with Acknowledgment $(ACK) = 901$.

What will be the updated Congestion Window (CWND) after this acknowledgement?

Your Answer: | Correct Answer: 1010 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Consider a TCP (Transmission Control Protocol) connection with the following parameters:

- Maximum Segment Size (MSS) = 1000 bytes
- Initial Congestion Window (CWND) = 1000 bytes
- Slow-Start Threshold (SSTHRESH) = 4000 bytes
- All data up to byte 6000 has been acknowledged (so the next expected byte is 6001).

Now the sender continues transmitting. After receiving the ACK for the 7th MSS, TCP updates its window. What will be the new CWND ?

Your Answer: | Correct Answer: 4250 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider a TCP connection implementing congestion control. At some instant, a timeout occurs when the congestion window (cwnd) is 32 KB . The maximum segment size (MSS) is 2 KB . The sender always has data to transmit, and ACKs are never lost.
After the timeout:

- The congestion threshold (ssthresh) is set to half of the current congestion window.
- The congestion window is reset to 1 MSS.
- The sender resumes with a slow start until cwnd reaches ssthresh and then continues in congestion avoidance.

Under these conditions, how many round-trip times (RTTs) are required for the congestion window to grow back to 32 KB ?

A. 8 RTTs

B. 11 RTTs

C. 14 RTTs

D. 16 RTTs

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A TCP sender is transmitting a 402 KB file using congestion control. The parameters are:

- ssthresh $= 64\ \mathrm{KB}$
- cwnd $= 2\ \mathrm{KB}$ initially (slow start)

- $\text{MSS} = 2\ \text{KB}$
- Receiver's window size $= 70\ \text{KB}$
- Sender always has data, and ACKs are never lost

By the time the 9th RTT is completed, how much data has the receiver acknowledged?

A. 192 KB

B. 260 KB

C. 330 KB

D. 400 KB

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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